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# Deep Residual Learning for Academic Document Processing

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## Abstract
Deep neural networks have revolutionized document processing. In this paper, we present an end-to-end asynchronous mic

## 1. Introduction
Academic research papers from publishers such as IEEE and Springer present unique layout challenges. Traditional machi

### 1.1 Problem Formulation
Let D be an input academic document consisting of N pages. Each page contains textual glyphs, formulas, and structural bo

## 2. Mathematical Modeling and Attention Mechanism

The self-attention mechanism operates on query (Q), key (K), and value (V) matrices computed from input tokens:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

where $d_k$ represents the dimensionality of key vectors. Multi-head attention extends this:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

### 2.1 Optimization and Loss Function
We optimize our objective using cross-entropy loss with AdamW optimizer:

$$\mathcal{L}_{CE} = -\sum_{i=1}^{M} y_i \log(\hat{y}_i) + \lambda \|\theta\|^2$$

where $\lambda = 0.01$ is the weight decay regularization parameter.

## 3. Experimental Results and Performance Analysis

Table 1 presents the comparative benchmark of translation models across standard academic datasets:

| Model Architecture | Parameters | BLEU Score | Latency (ms) | Throughput (tok/s) |
|---|---|---|---|---|
| CTranslate2 NLLB-200 | 600M | 34.2 | 243.2 | 128.5 |
| Qwen2.5-7B Instruct | 7.6B | 41.8 | 412.0 | 85.2 |
| LLaMA-3.1-8B Instant | 8.0B | 40.5 | 294.1 | 142.0 |
| Gemini 2.5 Flash | Large | 43.1 | 680.5 | 110.4 |

### 3.1 Ablation Study on Glossary Injection
When domain-specific glossary is injected into system prompts, the specialized terminology accuracy increases from 68.4%

## 4. Conclusion and Future Directions

In this work, we demonstrated that hierarchical parent-child chunking combined with Markdown-intermediate representation

## References

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